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Adjustable forehead support for a mask

Abstract

Forehead support adjuster for a mask, wherein the forehead support adjuster comprises at least a forehead support, consisting of support arm and cushion holder, a connection element, a rail with at least one guide groove, and a slide element, and wherein the slide element has at least one slide plate, at least one tenon and a guide element, wherein the at least one tenon is mounted rotatably and slidably in the at least one guide groove of the rail, and wherein the slide element is movably connected to the support arm. The forehead support adjuster is configured such that a movement of the slide element along the rail leads to a pivoting of the forehead support about at least one pivot axis D.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims priority under 35 U.S.C. § 119 of German Patent Application No. 102021001836.2, filed Apr. 9, 2021, the entire disclosure of which is expressly incorporated by reference herein.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The invention relates to an adjustable forehead support for a mask.

2. Discussion of Background Information

(3) To deliver respiratory gas from a ventilator to a patient, masks generally represent the interface between patient and ventilator. In addition to a gas-tight fit of the mask on the face of the patient, another important criterion is that the mask fits comfortably and precisely.

(4) In addition to a mask cushion resting around the nose and/or mouth of the patient, the orientation on the face also plays an important role. The forehead support also affords a possibility of influencing the fit of the mask. Various forehead supports are known from the prior art, but they are often non-adjustable, move only along one direction or have a bulky and/or complicated construction.

(5) It would therefore be advantageous to have available an adjustable forehead support which permits a secure and comfortable fit of the mask.

SUMMARY OF THE INVENTION

(6) In a first aspect, the invention relates to a forehead support adjuster for a mask, wherein the forehead support adjuster comprises at least a forehead support, consisting of support arm and cushion holder, a connection element, a rail with at least one guide groove, and a slide element, and wherein the slide element has at least one slide plate, at least one tenon and a guide element, wherein the at least one tenon is mounted rotatably and slidably in the at least one guide groove of the rail, and wherein the slide element is movably connected to the support arm. The forehead support adjuster is characterized in that the forehead support adjuster is configured such that a movement of the slide element along the rail leads to a pivoting of the forehead support about at least one pivot axis D.

(7) In some embodiments, the forehead support adjuster is characterized in that at least one inner wall is arranged on the slide plate, wherein the at least one tenon and the at least one guide element are arranged on the inner wall.

(8) In some embodiments, the forehead support adjuster is characterized in that at least two notches are arranged in the rail, and a latching lug is arranged on at least one tenon, which latching lug can latch into the notches and can thereby fix a pivoting position.

(9) In some embodiments, the forehead support adjuster is characterized in that the slide plate has, at two mutually opposite sides, respective wings which are substantially perpendicular to the slide plate.

(10) In some embodiments, the forehead support adjuster is characterized in that the wings have at least in each case two subsidiary wings, wherein an interspace is present between the respective subsidiary wings.

(11) In some embodiments, the forehead support adjuster is characterized in that the support arm has at least one carrier, which is rotatably connected to the connection element via at least one pin.

(12) In some embodiments, the forehead support adjuster is characterized in that discrete markings for marking the pivoting position are arranged on at least one side of the at least one carrier, wherein the discrete markings are arranged such that at least one marking can be seen in the interspace between the subsidiary wings when the latching lug fixes the corresponding pivoting position.

(13) In some embodiments, the forehead support adjuster is characterized in that the rail and the connection element are integrated in the mask body.

(14) In some embodiments, the forehead support adjuster is characterized in that the support arm has at least two carriers, wherein the carriers are arranged parallel to each other and are connected to each other via connections.

(15) In some embodiments, the forehead support adjuster is characterized in that the connections, together with the carriers, form a free space through which the slide element is at least partially plugged.

(16) In some embodiments, the forehead support adjuster is characterized in that at least two inner walls are arranged on the slide plate and are arranged substantially perpendicular to the slide plate, and wherein tenons and the guide elements are arranged at the edges that lie opposite the slide plate.

(17) In some embodiments, the forehead support adjuster is characterized in that the carriers are clamped between the slide plate and the guide elements.

(18) In some embodiments, the forehead support adjuster is characterized in that the carriers of the support arm have at least one guide edge, along which the guide elements are guided.

(19) In some embodiments, the forehead support adjuster is characterized in that the guide elements and tenons are arranged at a distance J from each other.

(20) In some embodiments, the forehead support adjuster is characterized in that the guide edge and

a top edge of the guide groove are at a distance K from each other when the guide edge and the guide groove extend parallel to each other.
(21) In some embodiments, the forehead support adjuster is characterized in that the distance J is not equal to the distance K.
(22) In some embodiments, the forehead support adjuster is characterized in that the distance J does not change during the movement of the slide element.
(23) In some embodiments, the forehead support adjuster is characterized in that the guide groove has a curved or straight configuration, wherein a curved guide groove has a radius R**1**.
(24) In some embodiments, the forehead support adjuster is characterized in that the guide edge (**30**, **49**) has a curved or straight configuration, wherein a curved guide edge (**30**, **49**) has a radius R**2**.
(25) In some embodiments, the forehead support adjuster is characterized in that the radius R**1** is not equal to the radius R**2**, and the ratio R**2**:R**1** or R**1**:R**2** lies in a range from 0.5 to 0.99, preferably 0.7 to 0.99, more preferably 0.75 to 0.985.
(26) In some embodiments, the forehead support adjuster is characterized in that the radii R**1** and R**2** lie in a range between 7 cm and 15 cm, preferably between 9 cm and 11 cm.
(27) In some embodiments, the forehead support adjuster is characterized in that the forehead support can be pivoted by a maximum pivoting angle C, wherein the maximum pivoting angle C lies between 100 and 40°, preferably between 150 and 25°.
(28) In some embodiments, the forehead support adjuster is characterized in that the top edge of the guide groove lies on a circle K**1** with the radius R**1**, and the guide edge lies on a circle K**2** with the radius R**2**.
(29) In some embodiments, the forehead support adjuster is characterized in that the pivot axis D always has the same relative position, i.e., lying inside or outside, with respect to the circles K**1** and K**2** when the maximum pivoting angle C is not exceeded.
(30) In some embodiments, the forehead support adjuster is characterized in that the pivot axis D always lies outside the circle K**2** and always inside the circle K**1**.
(31) In some embodiments, the forehead support adjuster is characterized in that a sliding of the slide element along the rail in the direction of the cushion holder causes the support arm to approach the mask body.
(32) In some embodiments, the forehead support adjuster is characterized in that the support arm transitions at one end into the cushion holder, wherein, at the end of the support arm lying opposite the end with the cushion holder, a peg is arranged on the connection, via which peg a closure piece is connected captively to the mask.
(33) In some embodiments, the forehead support adjuster is characterized in that the cushion holder has a holder frame which, together with a holder center, forms a free space into which at least one cushion receiver protrudes from at least one side of the holder frame, in the direction of the holder center, and serves to receive a forehead cushion.
(34) In some embodiments, the forehead support adjuster is characterized in that the tenons and guide elements are arranged and designed such that a sliding of the slide element leads to a movement of the tenons along the guide groove and a movement of the guide elements along the guide edge, wherein at the same time the forehead support is pivoted about the pivot axis D and the slide element is rotated about the tenons.
(35) In a further aspect, the invention relates to a mask with a forehead support adjuster according to the invention, wherein a gas attachment is arranged between attachment and connection element.
(36) In some embodiments, the mask is characterized in that the gas attachment is an attachment for an additional oxygen supply.
(37) In some embodiments, the mask is characterized in that the gas attachment can be closed with a closure piece, wherein this closure piece is captively connected to the mask via a peg on the forehead support.

(38) It is to be noted that the features individually presented in the claims can be combined with one another in any desired, technically meaningful way and show further refinements of the invention. The description additionally characterizes and specifies the invention in particular in conjunction with the figures.
(39) It also is to be noted that an "and/or" conjunction used herein between two features, and linking them to each other, is always to be interpreted as meaning that in a first embodiment of the subject matter according to the invention only the first feature may be present, in a second embodiment only the second feature may be present, and in a third embodiment both the first and the second feature may be present.
(40) A ventilator is to be understood as any appliance which supports the natural breathing of a user or patient, which takes over the ventilation of the user or living being (e.g., patient and/or neonate and/or premature baby) and/or which serves for respiration therapy and/or influences the respiration of the user or patient in some other way. This includes by way of example, but not exclusively, CPAP and BiPAP appliances, anesthesia appliances, respiration therapy appliances, ventilators (for use in hospitals, in non-hospital environments or in emergencies), high-flow therapy appliances and coughing machines. Ventilators can also be understood as diagnostic appliances for respiration. Diagnostic appliances can generally be used to detect medical and/or respiratory parameters of a living being. These also include appliances that are able to detect and optionally process medical parameters of patients in combination with respiration or only in relation to respiration.
(41) Unless specifically stated otherwise, a mask can be understood as any peripheral conceived for interaction with a patient, in particular for therapeutic or diagnostic purposes. The mask can be a full-face mask, i.e., enclosing the nose and mouth, or a nose mask, i.e. a mask enclosing only the nose. Tracheal tubes and cannulas and so-called nasal cannulas can also be used as mask.
(42) The fixing of a pivoting position by means of one or more latching lugs signifies releasable fixing. In the fixed position, a force generally has to be applied in order to move the slide element out of the fixed position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The invention is described in more detail by way of example with reference to FIGS. **1** to **22**.
(2) In the figures there is shown,
(3) FIG. **1**: a respiratory mask in a perspective view
(4) FIG. **2**: a mask body with forehead support
(5) FIG. **3**: a perspective exploded view of a mask body with forehead support
(6) FIG. **4**: an exploded side view of a mask body with forehead support
(7) FIG. **5**: a front view of a mask body with a forehead support adjuster, in order to illustrate the plane of symmetry
(8) FIG. **6**: a longitudinal section through forehead support and mask body
(9) FIG. **7**: a front view of a mask body
(10) FIG. **8**: a side view of a mask body
(11) FIG. **9**: a front view of a forehead support
(12) FIG. **10**: a longitudinal section through a forehead support
(13) FIG. **11**: a detailed view of a support arm in the region of pins for connecting to the connection element
(14) FIG. **12**: a bottom view of a slide element
(15) FIG. **13**: a rear view of a slide element
(16) FIG. **14**: a longitudinal section through a slide element

(17) FIG. **15**: a side view of a slide element
(18) FIG. **16**: a cross section through a support arm
(19) FIG. **17**: a cross section through rail/support arm/slide element
(20) FIGS. **18**A and **18**B: a pivoting range of a forehead support
(21) FIG. **19**: a schematic representation of the position of the guide groove and of the guide edge with respect to the circles K**1** and K**2**
(22) FIGS. **20**A and **20**B: a schematic representation of the function of the forehead support adjuster
(23) FIGS. **21**A and **21**B: a schematic representation of the distances between tenons and guide elements
(24) FIGS. **22**A and **22**B: a schematic representation of the distances between guide groove and guide edge
(25) The coordinate systems shown in some of the figures serve primarily to illustrate the orientation of the view; unless otherwise stated, the x, y and z axes describe the same direction in each case, i.e., the direction designated by the x axis in one figure corresponds to the direction of the x axis of another figure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

(26) The particulars shown herein are by way of example and for purposes of illustrative discussion of the embodiments of the present invention only and are presented in the cause of providing what is believed to be the most useful and readily understood description of the principles and conceptual aspects of the present invention. In this regard, no attempt is made to show details of the present invention in more detail than is necessary for the fundamental understanding of the present invention, the description in combination with the drawings making apparent to those of skill in the art how the several forms of the present invention may be embodied in practice.
(27) In the descriptions, it is assumed that the mask body as such has a fixed position and that the forehead support is moved by the forehead support adjuster. It is to be noted at this point that, conversely, the forehead support can be regarded as fixed and the mask body is moved by the forehead support adjuster. Ultimately, the forehead support adjuster provides a relative movement between mask body and forehead support.
(28) FIG. **1** shows an exemplary embodiment of the forehead support adjuster according to the invention, consisting at least of forehead support **5**, slide element **10**, rail **17** (not visible in FIG. **1**) and connection element **16** (not visible in FIG. **1**), on a mask **1** which is designed, for example, for use with a ventilator. For this purpose, for example, a hose attachment **4** is connected to the mask body **2**, and a hose system through which, for example, respiratory gas is conveyed to and from the patent can be connected to the hose attachment **4**. The attachment **14** is, for example, configured such that different types of hose attachments **4** can be connected to the mask **1**. In some embodiments, the attachment **14** also comprises exhalation structures through which respiratory gas can escape.
(29) The forehead support adjuster serves to adapt the position of the forehead support **5** or the forehead cushion **11**/cushion holder **9** relative to the mask body **2**. According to the invention, the forehead support adjuster consists at least of the forehead support **5**, a slide element **10**, a rail **17**, in which the slide element **10** is mounted rotatably and slidably, and a connection element **16** via which the forehead support **5** is connected rotatably or pivotably to the mask body **2**. The position of the forehead cushion **11**/cushion holder **9** relative to the mask body **2** is adapted by pivoting the forehead support **5** about a pivot axis D which is defined by the connection of the forehead support **5** to the connection element **16** by pins **27**.
(30) In some embodiments, it is also possible to have a plurality of pivot axes. For example, the pins **27** in the connection element **16** can assume different positions, such that a respective pivot axis can be found in each of the different positions. Moreover, the tenons **25** of the slide element **10** become a pivot axis through a movement between the different positions in the connection element

16.
(31) The mask body **2** is generally shell-shaped with at least two openings, such that an inner side F and an outer side E of the mask body **2** can be distinguished. The inner side F lies in the inner space, i.e., in the shell-shaped mask body, which substantially encloses a volume. The outer side E of the mask body **2** accordingly lies outside the mask body.
(32) Moreover, at least one head harness **7** is connected to the mask body **2**. In the embodiment shown, the head harness **7** is for example designed in one piece, wherein a connection to the mask **1** is effected in the region of the mask body **2**, here for example via harness clips **8**, and configured via the cushion holder **9** of the forehead support **5**. In addition to the one-piece head harness **7** shown, it is also conceivable to have a multi-part head harness **7**. Thus, a part of the head harness **7** can be connected to the mask **1** only in the region of the mask body **2**, and a further part of the head harness **7** only via the forehead support **5** (for example via the cushion holder **9**).
(33) For a gas-tight fit of the mask **1** on the face of a user, a face cushion **3** is also connected to the mask body **2**. It will be noted at this point that in principle any type of face cushion **3** can be used. It is assumed here that a skilled person knows how the mask body **2** could accordingly be adapted to different face cushions **3** in terms of a connection to the mask body **2**. In some embodiments, the forehead support adjuster according to the invention is also designed to be arranged and used in a mask **1**, which constitutes a one-piece design of the mask body **2** with face cushion **3**.
(34) The forehead support **5** with forehead support adjuster is for example arranged externally, i.e., on the outer side E, on the mask body **2**. The forehead support **5** can be divided here into the support arm **6** and the cushion holder **9**. The cushion holder **9** serves for example to receive the forehead cushion **11** and optionally also the holder of a head harness **7**. The support arm **6** constitutes a connection between cushion holder **9** and the mask body **2**. The forehead support adjuster according to the invention allows the position of the cushion holder **9** to be adapted by pivoting of the forehead support **5** about a pivot axis D, which lies at the end of the support arm **6** opposite the cushion holder **9**.
(35) FIG. **2** shows a perspective view of an exemplary embodiment of the mask body **2** with the forehead support adjuster. The mask body **2** has, for example, a cushion attachment **12** for connection to a face cushion **3**, and an attachment **14**, for example for connection to a hose attachment **4**. In some embodiments, the attachment **14** can also be designed in such a way that, for example, a filter element can be attached, for example also without a hose system for delivery of respiratory gas. The part of the mask body **2** in which the attachment **14** is arranged can be regarded, for example, as a lower part of the mask. The mask body **2** usually has a mirror-symmetrical shape, with the mirror plane G running centrally through the mask body **2** (see FIG. **5**). In particular, functional and cosmetic details can deviate entirely from the symmetry, for example asymmetrically arranged guide lines, for example for locking a hose attachment **4** in the attachment **14**. The attachment **14** is for example arranged at the bottom, centrally in the mask body **2**, such that the plane of symmetry G runs centrally through the attachment **14**.
(36) Clip holders **13**, which can receive harness clips **8** with the head harness **7**, are arranged laterally on the outside E of the mask body. For example, the harness clips **8** are clipped into the clip holder **13**, wherein the harness clips **8** can rotate in at least one plane in the clip holder **13** and, if appropriate, also have a certain degree of freedom of movement. Alternatively (or in addition) to a combination of clip holder **13** and harness clips, other ways of holding the head harness **7** on the mask body **2** are also possible.
(37) A closure piece **15**, which closes an additional, optional gas attachment **18**, is for example arranged above the attachment **14**. The closure piece **15** is, for example, connected captively to the support arm **6**. The closure piece **15** serves to close the gas attachment **18** when the latter is not required. In some embodiments, this gas attachment **18** is not arranged in the mask body **2**, and therefore the closure piece **15** is not arranged.
(38) The forehead support **5** is, as part of the forehead support adjuster, arranged above the closure

piece **15** or the gas attachment **18** on the outside E of the mask body **2** but is not part of the mask body **2**. At one end of the support arm **6**, the forehead support **5** is connected pivotably to the mask body **2** via at least one connection element **16** of the mask body **2**. Arranged at the other end of the support arm **6** is the cushion holder **9**, which is designed on the one hand to receive at least one forehead cushion **11** and a head harness **7**. The head harness **7** is guided around the harness receiver **41**, for example through at least one gap **36** of the cushion receiver **9**, and fixed. For example, the head harness **7** can be fixed by a kind of hook-and-loop closure.

(39) The forehead support adjuster moreover comprises the slide element **10** which, by way of tenons **25**, is connected to the mask body **2** so as to be rotatable and slidable in a rail **17** and also connected slidably to the support arm **6**. By sliding of the slide element **10**, the forehead support **5** is pivoted about the pivot axis D, which lies in the pins **27** connected to the connection element **26**.

(40) An exemplary embodiment of the forehead support adjuster is shown, with the mask body **2**, in an exploded view in FIGS. **3** and **4**. FIG. **3** shows a perspective view, FIG. **4** a side view.

(41) Among other things, the following are formed and/or arranged on the outer side E of the mask body **2**: the rail **17** for the rotatable and slidable connection to the slide element **10**, a connection element **16** for the movable, rotatable connection to the support arm **6** of the forehead support **5**, and the gas attachment **18** which can be closed by the closure piece **15**. The rail **17** and/or the connection element **16** can, for example, be fitted onto the mask body **2** or can be part of the mask body **2**. If rail **17** and/or connection element **16** are part of the mask body **2**, they are materially connected to the mask body **2** and preferably produced in one piece. If the rail **17** and/or connection element **16** are fitted on, they are connected to the mask body for example by being plugged on, glued on and/or screwed on.

(42) The cushion holder **9** is arranged at one end of the support arm **6** of the forehead support **5**. The cushion holder **9** has a holder frame **19**, from which the harness holder **41** starts. Between the harness holder **41** and the holder frame **19**, there is a gap **36** through which the head harness **7** can be guided. Also formed on the cushion frame **19** are, for example, two cushion receivers **38** via which the pockets **40** of the forehead cushion **11** can be plugged in order to connect the forehead cushion **11** to the cushion holder **9**. Alternatively, the pockets **40** can also be designed as a tab or tabs, in which case the tabs are each pulled through one of the cushion receivers **38**.

(43) At the end opposite the cushion holder **9**, the support arm **6** is pivotably connected to the connection element **16**. The connection to the connection element **16** at the same time constitutes the pivot axis D about which the support arm **6** or the forehead support **5** is pivotable. The support arm **6** moreover has a free space **37** through which the slide element **10** is at least partially inserted. The extent of the free space **37** determines the distance by which the slide element **10** is able to slide. In addition to the support arm **6**, the slide element **10** is also connected to the rail **17**. The slide element **10** is rotatably and slidably mounted inside the rail **17**, such that a pivoting movement of the forehead support **5** is permitted by the sliding of the slide element **10**.

(44) Arranged on the closure piece **15** is, for example, a closure tab **21** with which the closure piece **15** can be connected captively to the mask **1** or the forehead support **5**. For this purpose, the closure tab **21** has a hole, for example, through which a peg **39** of the support arm **6** can be plugged. This peg **39** is located, for example, in an interspace **35** of the connection element **16**. The closure tab **21** is, for example, designed such that it can be placed around the connection **23** of the support arm **6** (see, for example, FIG. **9**).

(45) FIG. **5** shows the front view (along the x axis; FIG. **4**) of an exemplary embodiment of the mask body **2** with forehead support adjuster and forehead support **5**. It will be seen that the mask body **2** and also the forehead support **5** are substantially mirror-symmetrical. The plane of symmetry G runs centrally through the mask body **2** and the forehead support **5** and parallel to the plane spanned by the x axis and y axis (see FIG. **4**). It will be noted that there is generally no perfect symmetry, and the respective mirror images may differ in a number of details, for example the design of the rail **17** of the mask body **2**, which rail **17**, in some embodiments, has notches **33**

on one side in the guide groove **20**. Moreover, the attachment **14** for example can also be designed such that asymmetrical elements are formed or various elements are arranged asymmetrically, for example in the guiding of the closure lock. Said elements, which can have an asymmetry, serve at this point only for explanatory purposes and do not in any way constitute a complete list of possible asymmetries.

(46) FIG. **6** shows a longitudinal section through an exemplary embodiment of the mask body **2** and of the forehead support adjuster, comprising among other things the forehead support **5** and the slide element **10**. The section plane corresponds substantially to the plane of symmetry G; the section plane is slightly offset laterally (in the z direction; FIG. **5**) such that the section is not precisely central.

(47) The gas attachment **18**, which produces a connection between the interior of the mask body **2**, surrounded by the inner side F, and the exterior of the mask body **2**, is closed by the closure piece **15** in the view shown in FIG. **6**. The closure piece **15** is connected captively to the support arm **6**, wherein the tab **21** is guided around the connection **23** at one end of the support arm **6** and pulled over the peg **39** (not shown in FIG. **6**) which is arranged at the connection **23**. The gas attachment **18** serves, for example, for additional introduction of oxygen or other gases into the interior of the mask. The gas attachment **18** is arranged, for example, between attachment **14** and connection element **16**, as a result of which the gas attachment is located in the nose region when the mask **1** is being worn by a user.

(48) The connection **23** extends for example over the connection element **16**, to which the forehead support **5** is rotatably or pivotably connected via pins **27**. The pins **27** thus at the same time constitute the pivot axis D about which the forehead support **5** is pivoted.

(49) The rail **17** of the mask body **2** is arranged above the connection element **16** (y direction). The rail **17** has at least one guide groove **20** in which the tenons **25** of the slide element **10** are slidably and rotatably mounted. In some embodiments, the rail **17** has a wall **46** centrally in the z direction, such that a guide groove **20** is formed on both sides of the wall **46**. The rail **17**, or at least the guide groove **20**, has a curved shape, for example, and has a radius R**1**. In some embodiments, this radius R**1** substantially follows the shape of the mask body **2**. In some embodiments, the radius R**1** is independent of the shape of the mask. In some embodiments, the guide groove **20** is straight and/or follows a parabola and/or a free form. In some embodiments, the guide groove **20** can alternate between regions in which it is parabola-shaped, circularly curved and/or straight.

(50) In some embodiments, the guide groove **20** is designed such that the pivot axis D lies on the arc or circle K**1** with radius R**1**, which is described by the curved shape of the guide groove **20**. In some embodiments, the pivot axis D also lies within the circle K**1** with radius R**1**, on the circumference of which the guide groove **20** runs. The radius R**1** measures, for example, between 7 and 15 cm, preferably between 9 and 11 cm.

(51) The slide element **10** is connected movably to the mask body **2** via the tenons **25** which are mounted rotatably and slidably in the guide groove **20**. The slide element **10** is thus able to slide in the rail **17**, while at the same time a rotational movement is possible, for example in interaction with the forehead support **5**. The tenons **25** are for example formed on inner walls **32** at a corner. Each inner wall **32** is formed with just one tenon **25**, so that a rotatability in the rail **17** is possible. The tenons **25** lie for this reason on a common axis. The inner walls **32** start, for example, from a slide plate **26** which bears, for example, on the support arm **6** of the forehead support **5**. To provide better grippability of the slide element **10**, grip structures **24**, for example, are formed among other things on the slide plate. These grip structures **24** are designed, for example, as sleeper-like elevations on the slide plate **26**. Besides this design, further forms of the grip structures **24** are also possible, for example knobs, or in the form of a grip recess.

(52) In some embodiments, the slide element **10** only has an inner wall, for example with a tenon **25** on both sides and guide elements **29** on at least one side.

(53) The slide element **10** constitutes a binding member between forehead support **5** and mask

body **2**. The forehead support **5** is therefore connected to the mask body **2** both via the pins **27** in the connection element **16** and via the slide element **10**. By virtue of the fact that the slide element **10** is mounted rotatably in the rail **17**, but the slide plate **26** of the slide element **10** bears on the support arm **6** of the forehead support **5**, a pivoting of the forehead support **5** is possible only when the slide element **10** is moved in the rail **17**. A free pivoting of the forehead support **5**, i.e. independently of the slide element **10**, is prevented by the slide element **10**.

(54) At that end of the support arm **6** not connected to the connection element **16** via the pins **27**, the support arm **6** transitions into the cushion holder **9**. The distance by which the slide element **10** can be pushed is limited by the connection **22**. This also prevents a situation where the slide element **10** is pushed out of the rail **17** and the forehead support **5** is freely rotatable about the pivot axis D.

(55) FIG. **7** shows an exemplary embodiment of the mask body **2** with the rail **17** and the connection element **16** in a front view. The rail **17** is arranged in the upper part of the mask body **2** (along the y axis above the gas attachment **18**). The connection element **16** is arranged below the rail **17**. The connection element **16** is composed, for example, of two walls, these walls each having an opening into which the pins **27** are inserted. In some embodiments, the pins **27** can also be arranged on the connection element **16** on the mask body **2**, and the support arm **6** has the corresponding openings into which the pins are inserted. In addition, any rotatable types of connection between support arm **6** and mask body **2** are also conceivable. For example, both the support arm **6** and the connection element **16** can have openings through which a pin is inserted which rotatably fixes the forehead support **5** in the connection element **16**.

(56) Located between the walls of the connection element **16** there is, for example, an interspace **35** that provides room for receiving the peg **39**.

(57) A side view of an exemplary embodiment of the mask body **2** is shown in FIG. **8**. The rail **17** is at least partially integrated in the mask body **2**. For example, the rail **17** has a wall **46** and accordingly has two guide grooves **20** on each side of the wall **46**. Several notches **33**, at least two thereof, are formed in the wall **46**, at least on one side. Alternatively or in addition, the notches **33** can also be arranged on both sides of the wall **46**. In some embodiments, an alternative or supplementary arrangement of the notches **33** in the top edge **51** of the rail **17** or in the side of the rail **17** opposite the top edge **51** is also possible. The slide element **10** is guided along the guide groove **20** with the tenons **25**. A latching lug **34**, which is formed on at least one of the tenons **25**, can latch into the notches **33** of the guide groove **20** and thereby fix the set position of the forehead support **5**. The latching lug **34** is here arranged on the tenon **25** such that latching into the notches **33** is possible. If the notches **33** are arranged for example in the top edge **51** of the rail **17**, the latching lug **34** is formed for example on the corresponding side of the tenon **25**. In some embodiments, the guide groove **20** has no notches, and the tenons **25** are also designed without latching lug **34**. In some embodiments, latching points are also realized at other locations or not formed at all, such that the slide element **10** cannot latch in. The rail **17** or at least the guide groove **20** is curved and has a radius R**1**. In alternative or supplementary embodiments, the guide groove **20** has a linear shape or parabola shape and/or a combination between curved, straight and/or parabola-shaped.

(58) An exemplary embodiment of the forehead support **5** is shown in FIG. **9** in a front view. The forehead support **5** can be roughly divided into the support arm **6** and the cushion holder **9**, the support arm **6** transitioning at one end into the cushion holder **9**. The cushion holder **9** begins, for example, at the place where the substantially curved support arm **6** ends at an edge. In addition to a curved shape, the support arm **6** can also be straight and/or parabola-shaped. From this edge, the cushion holder **9** widens to the sides among other things. The cushion holder **9** is rotated through 90° with respect to the support arm **6**, as a result of which the front view of the forehead support **5** reveals what is substantially a T shape. For example, the cushion holder **9** thus has a width which is at least twice as great as the other dimensions. In some embodiments of the forehead support **5**, the

cushion holder **9** has other shapes. In the embodiment shown in the figures (at least FIGS. **1-5**, **8**, **9** and **18**), the cushion holder **9** describes a rectangle in a front view, although it can also adopt all possible other shapes, for example a square, round or polygonal shape or a free form.
(59) The cushion holder **9** serves on the one hand to receive the forehead cushion **11** and on the other hand also to receive the head harness **7**. The forehead cushion **11** is connected to the cushion holder **9**, for example via the cushion receivers **38**. In the exemplary embodiment of the forehead support **5**, the cushion receivers **38** extend from the holder frame **19**, parallel to the indicated z axis, inward into the free space **48** formed by the holder frame **19** and the holder center **47**. However, the cushion receivers **38** do not span the whole free space from holder frame **19** to holder center **47**, and instead they fill this free space **48** only partially. The cushion receivers **38** are designed, for example, as plates which at one edge are connected to the holder frame **19** or transition into it. The pockets **40** of the forehead cushion **11** can be pulled over these plates, as a result of which the forehead cushion **11** is fixed on the cushion holder **9**. In some embodiments, the cushion receivers **38** can also adopt other forms. For example, it is conceivable that the cushion receivers **38** are designed not as plates but as a plurality of rods, in which case the forehead cushion **11** has for example, instead of the pockets **40**, several eyelets or tabs which can be pulled over the rods. In some embodiments of the cushion holder **9**, in addition to the cushion receivers **38** which protrude inward from the holder frame **19** in the direction of the holder center **47**, there are also further cushion receivers which protrude from the holder center **47** into the free space **48**. For example, the forehead cushion to be fitted thereto has a corresponding number of pockets which are pulled over the cushion receivers in order to fix the forehead cushion on the cushion holder **9**.
(60) The harness receiver **41** forms, with holder frame **19**, a gap **36** through which the head harness **7** can be guided and placed around the harness receiver **41**. A harness receiver **41** issues in each case from the outer sides of the holder frame **19** (in the z direction). Between the harness receiver **41** and the holder frame **19**, a gap **36** forms through which the head harness **7** is guided and then wound around the harness receiver **41**. In the embodiment of the forehead support **5** shown in FIGS. **9** and **10**, the gap **36** results principally from the fact that the harness receiver **41** protrudes in particular in the x direction (coordinate system of FIG. **10**) from the holder frame **19**. In some embodiments, the harness receiver **41** can also issue mainly in the z direction from the holder frame **19**.
(61) The support arm **6** of the forehead support **5** has, for example, two carriers **28**, which are connected in the upper and lower region (according to the y axis) by the connection **23** and the connection **22**, respectively. The carriers **28** run parallel to each other, for example. By means of the carriers **28** and the edges of the connections **22** and **23**, a free space **37** is formed through which the slide element **10** is plugged onto the support arm **6**. The connection **22** here limits how far the slide element **10** can be pushed and in some cases serves also to prevent the slide element **10** from sliding out of the rail **17**.
(62) The connection **23** at the lower end of the support arm **6** has a gap **42**, for example. The closure tab **21**, for example, can be guided through this gap **42** when placed around the connection **23** and pulled over the peg **39**.
(63) A longitudinal section through the forehead support **5**, along the section edge X-X, is shown schematically in FIG. **10**. Arranged at one of the ends of the support arm **6** are the pins **27** which are inserted into the connection element **16** in order to produce a rotatable connection between forehead support **5** and mask body **2**. The pins **27** at the same time also constitute the pivot axis D about which the forehead support **5** can be pivoted or can rotate. At an edge **49** of the two carriers **28**, which are materially connected at one end by the connection **23** and at the other end by the connection **22**, respective guide edges **30** are formed which bear on the guide elements **29** of the slide element **10** and/or are clamped by the slide element **10** between slide plate **26** and guide elements **39**.
(64) The guide edges **30** have a curved shape, for example, and have a radius R**2** which, for

example, is equal to the radius R**1** of the guide groove **20**. For example, the circumference of the circle K**2** with radius R**2**, on which the guide edges extend, does not extend through the pivot axis D. For example, the pivot axis D lies outside the circle K**2** with radius R**2** on the circumference of which the guide edges **30** extend. In some embodiments, the pivot axis D also lies on the circumference of the circle K**2** with radius R**2**.
(65) In each case, the circle K**2** is not congruent with the circle K**1**; the circles K**1** and K**2** do not have a common center point. The circles K**1** and K**2** preferably lie such that they mutually intersect. The intersection points of K**1** and K**2** can lie in a continuation of the guide groove **20** or of the guide edges **30**.
(66) The pivot axis D can also lie, for example, at an intersection point of the two circles K**1** and K**2**. In some embodiments, the pivot axis lies inside the circle K**1** and outside the circle K**2**. FIG. **19** shows the circles more clearly in relation to pivot axis D, guide groove **20** and guide edge **30**.
(67) In some embodiments, the guide edges **30** have a radius R**2** which is smaller or greater than the radius R**1** of the guide groove **20** of the rail **17**. In these embodiments, the degree of pivotability of the forehead support **5** is defined in particular by the radius R**2**. In some embodiments of the forehead support adjuster, the pivotability of the forehead support is defined by the ratio of the radius R**1** to the radius R**2**, wherein the radius R**2** should be smaller or greater than the radius R**1**. In particular, the ratio between R**2** and R**1** defines to what extent the forehead support is pivotable. If R**1** is greater than R**2**, the ratio of R**2** to R**1** is for example in a range R**2**:R**1** of 0.5 to 0.99, preferably in a range of 0.7 to 0.99, more preferably between 0.75 and 0.985. The smaller the ratio of R**2** to R**1**, the greater the angle by which the forehead support **5** pivots upon movement of the slide element **10**. A movement of the slide element **10** leads, at a smaller ratio R**2**:R**1**, to a greater pivoting angle of the forehead support **5** than for the same movement of the slide element **10** at a greater ratio R**2**:R**1**. If R**2** is greater than R**1**, the ratio of R**1** to R**2** lies for example in a range R**1**:R**2** of 0.5 to 0.99, preferably in a range of 0.7 to 0.99, more preferably between 0.75 and 0.985. If the radii R**1** and R**2** are different, an intersection point of the respective circles K**1** and K**2** is not necessary for a pivotability of the forehead support, provided the circles K**1** and K**2** have no common center point.
(68) For example, the radius R**2** measures between 7 and 15 cm, preferably between 9 and 11 cm.
(69) In some embodiments, the guide edge **30** does not have a curved shape that forms the basis of a circle. For example, the guide edge **30** can extend in any desired curving shape, e.g., parabola-shaped, or can be straight. The choice of the course of the guide edge **30** is independent of the shape of the guide groove **20**.
(70) At the end of the support arm **6** opposite the pins **27**, the connection **22** transitions materially into the holder center **47** of the cushion holder **9**. From the holder center **47** there also extends the holder frame **19** from which, among other things, the harness receivers **41** and the cushion receivers **38** start. The harness receivers **41** each form, together with the holder frame **19**, a gap **36** through which the head harness **7** can be guided and placed around the harness receiver **41**.
(71) FIG. **11** shows in detail the end of the support arm **6** at which the pins **27** and the peg **39** are arranged. The carriers **28** of the support arm **6** are connected to each other via the connections **22**, **23** and delimit, together with the carriers **28**, the free space **37** which can receive the slide element **10**, by the inner walls **32** of the slide element **10** being plugged through the free space **37**. At those edges **49** of the carriers **28** that lie opposite the connections **22**, **23**, curved (or curve-shaped) guide edges **30** are formed, for example. These guide edges **30** are arranged internally, for example. In some embodiments, no extra guide edge **30** is formed on the carriers **28**, and instead the edges **49** of the carriers **28** themselves serve as guide edges **30**.
(72) The connection **23**, which interconnects the carriers **28** at the end at which the pins **27** are arranged, has a peg **39** for example. The peg **39** extends for example parallel to the carriers **28** from the connection **29** in the direction of the mask body **2** and is received by the interspace **35** of the connection element **16**. The peg **39** can be guided for example through the hole of the closure tab

21 in order to produce a permanent connection between closure piece **15** and mask **1**. In some embodiments, the mask body **2** does not have a gas attachment **18**, so that the closure piece **15** is also not needed. In these embodiments, the peg **39** can also not be formed on the support arm **6**.
(73) The pins **27** are formed at one end of the carriers **28** or of the support arm; the end where the pins **27** are arranged lies opposite the end where the support arm **6** transitions into the cushion holder **9**. The pins **27** protrude for example perpendicularly from the carriers **28**. For example, the pins **27** are directed inward, i.e. protrude toward each other, but do not touch each other. In some embodiments, the pins **27** are arranged externally on the carriers **28** and are oriented toward the outside, i.e. point away from each other. Instead of the pins, holes can also be formed. If holes are arranged instead of pins **27** at the end of the carriers **28**, then the connection element **16** can for example have corresponding pins over which the holes of the support arm **6** are fitted. Alternatively or in addition, both the connection element **16** and the support arm **6** can each have holes. In this case, a rod/a peg is inserted through the holes for the connection. The pivot axis D of the forehead support **5** is defined by the pins **27** of the support arm **6**.
(74) An exemplary embodiment of the slide element **10** is shown schematically in FIGS. **12** to **15**. The coordinate systems indicated in FIGS. **12** to **15** relate exclusively to FIGS. **12** to **15** and do not correspond exactly to the coordinate systems of the other figures.
(75) FIG. **12** shows a bottom view of the slide element **10**, i.e. the underside of the slide plate **26**. On the underside of the slide plate **26**, two inner walls **32** are formed which are substantially perpendicular to the slide plate **26** and parallel to each other. The guide elements **29** and the tenons **25** are arranged on the inner walls **32**, preferably at the edge of the inner walls **32** lying opposite the slide plate **26**. The tenons **25** are arranged for example at one of the corners of the inner walls **32**, for example directed inward, i.e. pointing toward each other. A latching lug **34** is additionally formed on at least one of the tenons **25** and can latch into the notches **33** of the rail **17** in order to limit or fix the sliding movement of the slide element **10**, such that discrete adjustment steps of the forehead support **5** can be chosen alternately.
(76) In some embodiments, the slide element **10** has only one inner wall, for example with a respective tenon **25** on both sides and guide elements **29** on at least one side. In some embodiments, it is also possible that a third inner wall **32** is arranged in parallel between the two inner walls **32**. For example, this inner wall can have additional guide elements **29** and/or tenons **25**. Accordingly, further rails **17** or guide grooves **20** and/or also guide edges **30** would also have to be arranged for this purpose.
(77) The guide elements **29** are formed at the same edge of the inner walls **32** where the tenons **25** are arranged. For example, a guide element **29** is arranged at each corner. The guide elements **29** are for example arranged such that they point outward from the inner wall **32**, i.e. are oriented away from each other. At least one side, preferably the side pointing to the slide plate **26**, is substantially flat, the surface thus shown being in some embodiments curved with a radius R**3**. In some embodiments, the radius R**3** corresponds for example to the radius R**2** of the guide edge **30**. In some embodiments, the guide elements **29** are configured such that they are adapted to the course of the guide edge **30**. In some embodiments, the guide elements **29** are round, such that there is a minimal contact surface between guide element **29** and guide edge **30**.
(78) In some embodiments, the guide elements **29** are oriented inward and formed on the inner side of the inner walls **32**. Accordingly, in some embodiments, the tenons **25** are arranged on the outer side of the inner walls **32** and are directed outward. In some embodiments, tenons **25** and guide elements **29** are arranged on the same side of the respective inner wall **32**. In this case, the guide elements **29** should be offset upward, in the direction of the slide plate **26**, and the tenon **25** arranged such that the support arm **6** can be arranged and pivoted above the rail **17**.
(79) Instead of two individual guide elements **29** per inner wall **32**, in some embodiments a continuous guide element can also be formed per inner wall **32** and extends over the whole or at least most (>50%) of the length (y direction in FIG. **12**) of the inner wall **32**.

(80) On the outer (z direction) edges of the slide plate **26**, the slide plate transitions into the wing elements **31**. Grip structures **24** are arranged on the wing elements **31**, which in some embodiments transition into at least two subsidiary wings **44**, said grip structures **24** allowing a better grip of the slide element **10**. The wing elements **31** are substantially perpendicular to the slide plate **26** and extend parallel to the inner walls **32**.

(81) A front view (along the y axis in FIG. **12**) of an exemplary embodiment of the slide element **10** is shown in FIG. **13**. Grip structures **24** for example, which allow an improved grip of the slide element **10**, are arranged on the slide plate **26** and on the sides of the wing elements **31**. Two mutually parallel inner walls **32** start out from the slide plate **26**. The inner walls **32** are substantially perpendicular to the slide plate **26**; in some embodiments the slide plate **26** has a curvature. The wing elements **31**, into which the slide plate **26** transitions at the sides, extend parallel to the inner walls **32**.

(82) The guide elements **29** are arranged, pointing outward, both in the front region and in the rear region of the inner walls **32**. The guide elements **29** protrude perpendicular to the inner walls **32**. For example, the guide elements **29** are flat on the side facing the slide plate **26**, such that the guide edge **30** can bear thereon. The guide elements **29** are for example angled in a downward direction (x direction), thereby permitting easier insertion into or through the free space **37** of the forehead support **5**. The carriers **28** of the support arm **6** are for example clamped movably between slide plate **26** and guide elements **29**. A movement of the slide element **10** along the support arm **6** thus remains possible, while an inadvertent release of support arm **6** and slide element **10** is prevented. Moreover, pivoting of the forehead support **5** is not possible without simultaneous rotational movement of the slide element **10**; the orientation of the slide element **10** with respect to the support arm **6** always remains the same.

(83) An inwardly directed tenon **25** is arranged in the rear region of each of the inner walls **32** and can be received in the guide groove **20** of the rail **17**. The tenons **25** are designed such that they are mounted rotatably and slidably in the guide groove **20**. At least one of the tenons **25** has a latching lug **34** which can latch into the notches **33** of the rail **17** in order to fix the setting of the forehead support with respect to an inadvertent adjustment.

(84) FIG. **14** shows a section through an exemplary embodiment of the slide element **10** along the section edge Y-Y (FIG. **12**). The slide plate **26** is slightly curved/arcuate, for example, wherein for example an arc is described which corresponds approximately to the radius R**2** of the guide edge **30**. The radius of the slide plate **26** can deviate slightly (+/−10%) from the radius R**2**. In some embodiments, the slide plate **26** is also plane/flat. Grip structures **24** are for example arranged on the slide plate **26**.

(85) In some embodiments, the slide plate **26** is for example shaped corresponding to the carriers **28**. If, for example, the carriers **28** extend in a straight line or in a parabola shape, the slide plate **26** likewise has a straight or parabola-shaped form; however, at least the underside of the slide plate **26**, in the regions where the latter bears on the carrier **28**, substantially follows the shape of the carrier **28**.

(86) The inner wall **32** extends substantially perpendicular to the slide plate **26**, on the side of the slide plate **26** lying opposite the grip structures **24**. The view shown in FIG. **14** is directed to the inner side of one of the inner walls **32**. In one of the corners of the inner wall **32**, at the edge lying opposite the slide plate **26**, a tenon **25** is formed which is received in the guide groove **20** of the rail **17** in order to achieve rotatable and slidable mounting of the slide element **10** in the rail **17**. For example, a latching lug **34** is formed on the tenon **25** and is able to latch into the notches **33** of the guide groove **20** or of the rail **17**.

(87) FIG. **15** shows a side view of an exemplary embodiment of the slide element **10**. The slide plate **26** transitions for example into the wing **31** which runs parallel to the inner wall **32**. The wing **31** has, for example, two subsidiary wings **44**, on each of which a grip structure **24** is arranged. An interspace **43** is located between the subsidiary wings **44**. In some embodiments, discrete markings

(for example a numbering and/or value indications, e.g. pivoting angle) for the pivoting positions and/or for marking the pivoting position are arranged, for example printed, on the outer side of the support arm **6**. A pivoting position can for example be characterized in that the slide element **10** in the corresponding position is able to latch with the latching lug **34** in a notch **33**. These markings are for example arranged such that the marking corresponding to the pivoting position can in each case be seen in the interspace **43** between the subsidiary wings **43**.
(88) The two guide elements **29** shown in FIG. **15** are for example arranged at a slight angle to each other. The guide elements **29** angled relative to each other correspond substantially to the arc-shaped/curved profile of the guide edge **30**. In some embodiments, the guide elements **29** themselves are curved, such that the described arc has a radius R**3** which substantially corresponds to the radius R**2** of the guide edge **30**.
(89) The guide elements **29** are for example formed with an elongate extent (in y direction of FIG. **15**). In some embodiments, the guide elements **29** can also have a substantially round shape in a plan view (along the viewing axis in FIG. **15**).
(90) FIG. **16** shows a section through an exemplary embodiment of the support arm **6** in an exemplary arrangement with a rail **17** on the mask body **2** and with an inserted slide element **10**. The section plane through the support arm **6** is set directly after the start of the connection **22** in the attachment to the free space **37**. In the view shown, the slide element **10** is for example located with the latching lug **34** on one of the tenons **25** in the last notch of the rail **17**, that is to say in the position farthest from the connection element **16**. A further movement of the slide element **10** away from the connection element **16** is prevented by the connection **22**, on which the slide element **10** abuts with the inner walls **32**. In this setting, the forehead support **5** lies closest to the mask body **2**. If the slide element **10** is pushed in the direction of the connection element **16**, the support arm **6** is pivoted about the pivot axis D, which for example is defined by the pins **27** in the connection element **16**.
(91) The tenons **25** of the slide element **10** are rotatably and slidably mounted in the guide groove **20** of the rail **17** and can thus be moved along the rail **17** and rotated in the process. The carriers **28** of the support arm **6** bear for example with the guide edge **30** on the guide elements **29** of the slide element **10** or are clamped between the slide plate **26** and the guide elements **29**. Depending on the direction of movement of the slide element **10**, either the slide plate **26** presses from above on the carriers **28** or the guide elements **29** press from below on the guide edge **30** in order to effect a pivoting movement of the forehead support **5** about the pivot axis D. The wing elements **31** of the slide element **10** are arranged on the slide element **10** such that they are arranged externally (in the z direction) on the sides of the carriers **28**. The inner walls **32** of the slide element **10** are located for example on the side of the carriers **28** opposite the wing elements **31**.
(92) If the slide element **10** is pushed in the direction of the connection element **16**, the slide element **10** is guided along the guide groove **20** of the rail **17**. Since the circles of the guide edge **30** of the support arm **6** and of the guide groove **20** do not have a common center point or the contours of the guide groove **20** and of the guide edge **30** do not run parallel, the guide elements **29** come into contact with the carriers **28** or the guide edge **30** during the movement. Since the support arm **6** is mounted pivotably in the connection element **16**, a pivoting of the support arm **6** can take place, as a result of which it yields to the pressure of the guide elements **29** of the slide element **10**. The carriers **28** in turn exert a pressure on the slide plate **26** through the pivoting movement of the forehead support **5**. In order to yield to this pressure, the whole slide element **10** rotates in the guide groove **20**. In the sliding movement of the slide element **10** in the direction of the connection element **16**, the slide element **10** and the forehead support **5** thus pivot or rotate simultaneously. The axis of pivoting or axis of rotation of the slide element **10** runs through the two tenons **25**.
(93) FIG. **17** shows a section through an exemplary embodiment of the forehead support adjuster consisting of slide element **10**, forehead support **5** and rail **17**. The carriers **28** are clamped between slide plate **26** and guide elements **29**, the guide edge **30** bearing on the guide elements **29**. In some

embodiments, no special guide edge **30** is formed on the carriers **28**, and therefore the lower edge of the carrier **28** itself constitutes the guide edge **30**.

(94) The tenons **25** of the slide element **10** run in the guide groove **20** of the rail **17**. On one of the tenons **25**, a latching lug **24** for example is arranged which is able to latch in the notches **33** of the rail **17** and thereby prevents an inadvertent sliding movement of the slide element **10** and fixes the forehead support **5** in a pivoting position. At the sides of the carriers **28**, the wing elements **31** and the inner walls **32** extend perpendicularly from the slide plate **26**.

(95) For example, the tenons **25** and the guide elements **29** are arranged at the same height (x direction in FIG. **17**) on the inner walls **32** of the slide element **10**. In some embodiments, the tenons **25** and the guide elements **29** can also be arranged at different heights. In some embodiments, the arrangement on the outer side (guide elements **29**) and inner side (tenons **25**) of the inner walls **32** may also be different. In some embodiments, the forehead support adjuster is for example constructed such that the carriers **28** run with the guide edge **30** centrally, above the rail **17**. In this case, guide elements **29** and tenons **25** would both be arranged on the inner sides of the inner walls **32**. The tenons **25** are in each case to be arranged on that side of the inner walls **32** on which the rail **17** with the guide groove **20** runs. The guide elements **29** are in each case to be arranged on the side on which the guide edge **30** runs.

(96) The rail **17** is for example connected materially to the mask body **2**. For example, mask body **2** and rail **17** are produced in one piece. In some embodiments, however, the rail **17** can also be produced as an extra and placed onto the mask body **2** and connected to the mask body **2** by plugging, clamping, adhesive bonding and/or screwing or by similar connection methods.

(97) The illustrative pivoting range of the forehead support **5** is shown in FIGS. **18**A and **18**B. In FIG. **18**A, the slide element **10** is in the latching position which lies closest to the connection element **16** and thus to the pivot axis D. The latching lug **34** is thus latched in the notch **33** which lies closest to the connection element. In this position, a plane A can for example be defined between cushion holder **9** and forehead cushion **11**, said plane A being defined substantially by the surface of the cushion holder **9**. This plane A can be shifted in parallel so that the pivot axis D lies on the plane A.

(98) In FIG. **18**B, the slide element **10** is located in the latching position which lies farthest away from the connection element **16** and therefore the pivot axis D. Considering here the plane B, which for example can be defined on the basis of the surface of the cushion holder **9**, it is rotated relative to the plane A by the angle C. The angle C constitutes for example the maximum pivoting angle of the forehead support **5** about the pivot axis D. For example, the maximum pivoting angle C of the forehead support **5** lies in a range from 10° to 40°, preferably between 15° and 25°.

(99) The maximum pivoting angle C here indicates the change of the angle that arises between the respective first and last pivoting positions. For example, the pivoting angle at which the slide element **10** sits in the latching position closest to the connection elements **16** can be defined as 0°. In the latching position farthest from the connection element **16**, the pivoting angle then assumes the value of the maximum pivoting angle C.

(100) The maximum pivoting angle C depends on a number of factors. In particular, the distance by which the slide element **10** can be moved influences the maximum pivoting angle. The longer this distance is chosen to be, the greater the pivoting angle C. The distance by which the slide element **10** can be moved in the guide groove **20** is for example between 2 cm and 10 cm, preferably between 2.5 cm and 5 cm, more preferably between 2.5 and 4 cm.

(101) Moreover, the maximum pivoting angle is defined by the ratio of the radius R**2** of the guide edge **30** to the radius R**1** of the guide groove **20**. For example, the maximum pivoting angle C increases, with a constant distance for the slide element **10**, when the ratio R**2**:R**1** decreases. If the distance by which the slide element **10** can be moved is lengthened, the maximum pivoting angle C increases at constant ratio R**2**:R**1**.

(102) During the pivoting of the forehead support **5** about the pivot axis D, the positioning of the

cushion holder **9** changes and thus also the positioning of the forehead cushion **11** relative to the mask body **2**. The orientation of the forehead cushion **11** also changes by the angle C. As has been described for FIGS. **18**A and **18**B, the forehead cushion moves counter to the x direction and counter to the y direction during the pivoting of the forehead support **5**. From FIG. **18**A to FIG. **18**B, the forehead support **5** is basically guided closer to the mask body **2**.
(103) If the slide element **10** is moved along the rail **17** in the direction of the cushion holder **9**, for example from the position in FIG. **18**A to the position in FIG. **18**B, the support arm **6** approaches the mask body **2**, until the maximum pivoting position is reached.
(104) Assuming that the mask is resting on a face, and the forehead support **5** is fixed with a head harness on the forehead of the user, an adjustment of the forehead support **5** would effect a change of the positioning of the mask body **2**. For example, by a movement of the slide element **10** to the latching position farthest from the connection element **16**, the upper region of the mask body **2** is pressed away from the user's face.
(105) An example of the relative position of the circles K**1** and K**2**, with the guide groove **20** and guide edge **30** extending on the respective circumference, is shown schematically in FIG. **19**. The position shown is for example only valid for one pivoting position. The circles K**1** and K**2** have radii R**1** and R**2**, which are of the same magnitude for example. The center points of the circles K**1** and K**2** are offset relative to each other, such that the circles intersect. The guide groove **20** runs along a partial segment of the circumference of the circle K**1**. The guide edge **30** runs along a partial segment of the circumference of the circle K**2**. For example, guide edge **30** and also guide groove **20** have an end in the region of the point of intersection of the circles K**1** and K**2**. In the exemplary embodiment shown, the pivot axis D, which is defined for example by the pins **27**, is located inside the circles K**1** and K**2**. During the movement of the slide element **10**, the circle K**2** turns about the pivot axis D while the circle K**1** remains fixed. In this way, the guide edge **30** pivots relative to the guide groove **20** about the pivot axis D. In some embodiments, the pivot axis D can be arranged freely. Thus, an arrangement outside K**1** but inside K**2**, an arrangement outside both K**1** and K**2**, and also an arrangement on the circumference of K**1** and/or K**2** are possible. During the pivoting of the carrier **28** about the pivot axis D, the position of the circles K**1** and K**2** relative to each other also changes at the same time. By virtue of the fact that the circles K**1** and K**2** come into a congruent position or the circles run parallel to each other, it is for example possible to prevent a situation where the distance J between the guide elements **29** and the tenons **25** does not correspond to the distance K between the top edge **51** of the guide groove **20** and the guide edge **30**, when the guide groove **20** and the guide edge **30** are set parallel to each other by a pivoting about the pivot axis D.
(106) Whether the pivot axis D is inside or outside the circle K**2** which is rotatable about the pivot axis D depends primarily on the current pivoting position. However, in connection with the maximum pivoting angle C, it is possible to ensure that the pivot axis D always lies outside or inside the circle K**2**. It is thus also possible to ensure, for example, that during a pivoting movement the position of the pivot axis D relative to the circle K**2** alternates from inside to outside and vice versa.
(107) Also in the case of a straight or parabola-shaped course of the guide groove **20** and of the guide edge **30**, the distance J between the guide elements **29** and the tenons **25** should not correspond to the distance K between the top edge **51** of the guide groove **20** and of the guide edge **30** in a parallel setting, since otherwise no pivoting movement can be achieved through the sliding of the slide element **10**. The distance J between the guide elements **29** and the tenons **25** is preferably greater than the distance K between the top edge **51** of the guide groove **20** and of the guide edge **30**, when these extend parallel to each other through a pivoting position.
(108) The function of an exemplary embodiment of the forehead support adjuster is shown schematically in FIG. **20**, in which only the tenons **25**, the slide plate **26** and the guide elements **29** of the slide element **10** are shown. For example, an embodiment is shown in which guide groove **20**

and also guide edge **30** and carrier **28** have a curved shape and follow a circumference of a circle. The radius R**2** is for example equal to the radius R**1**, the circles K**1** and K**2** intersect for example, and the pivot axis D lies for example outside circle K**2** and inside circle K**1**. The profile of the circles K**1** and K**2** is indicated for example by the continuations designated K**1** and K**2**. The slide element **10** comprises, among other things. the slide plate **26**, the tenons **25** and the guide elements **29**, which are arranged at a fixed distance to one another. This is achieved, for example, by the slide element **10** being produced in one piece.
(109) If the slide element **10** is moved in the direction of the pivot axis D, which is defined for example by the position of the pins **27**, the tenons **25** are guided along the guide groove **20**. The guide elements **29** press on the guide edges **30** of the carriers **28** (part of the support arm **6**). In this way, the support arm **6** is pivoted about the pivot axis D. The carrier **28** is clamped between the slide plate **26** and the guide elements **29**, the arrangement of the slide plate **26** and of the guide elements **29** being adapted to the contour of the carriers **28**. In order to further follow this contour during the pivoting of the carriers **28**, the guide elements **29** and the slide plate **26** move along the carrier. In the process, the slide element **10** at the same time rotates about the tenons **25**.
(110) If the slide element **10** is moved away from the pivot axis D, the slide plate **26** for example presses on the carriers **28**. As a result of this, a rotational movement of the slide element **10** about the tenons **25** takes place.
(111) The slide element **10** is configured to follow the profile of the circle K**1**, i.e. of the guide groove **20**, and also the profile of the circle K**2**, i.e. of the guide edge **30**, and/or the carrier **28**. The two circles are not congruent, such that, for example during a movement of the slide element **10** along the guide groove **20**, a change of the orientation of the slide element **10** by a rotation about the tenons **25** is necessary at the same time, so that the slide element **10** is able to follow both profiles, of K**1** and K**2**.
(112) In some embodiments, a movement of the slide element **10**, i.e. a movement with simultaneous rotation about the tenons **25**, can for example also be achieved when the forehead support **5** is pivoted about the pivot axis D. Depending on the direction of pivoting, the carrier **28** applies pressure to the slide plate **26** or the guide elements **29**, which then seek to follow the contour of the carriers **28** or of the guide edge **30**.
(113) In the position A in FIG. **20**, the guide edge **30** has at one end a distance H from the guide groove **20**. If the slide element **10** is pushed in the direction of the pivot axis D (position B in FIG. **20**), the support arm **20** pivots, as a result of which the distance of the guide edge **30** from the guide groove **20** increases to the distance M. By the movement of the slide element **10** in the direction of the pivot axis D, the distance between carrier **28** and guide groove **20** increases, i.e. also from the mask body **2**.
(114) FIG. **21**A shows a greatly simplified, schematic view of an exemplary embodiment of the slide element **10**. Here, reference is made in particular to the distance J between the top edge (in x direction) of the guide elements **29**, which are arranged at the same distance J from the tenon **25** and are straight, and the uppermost point of the tenon **25**. The distance J is preferably not equal to the distance K between the top edge **51** of the guide groove and the guide edge **30** when, by pivoting, they are brought to a position in which guide groove **20** and guide edge **30** run parallel to each other or, in the case of curved shapes of identical radius, are congruent.
(115) FIG. **21**B shows an individual guide element **29** which for example is curved and is designed with the radius R**3**, where the radius R**3** corresponds for example to the radius R**2** of the guide edge **30**. In such an embodiment, the distance J is between the highest point in the x direction of the tenon **25** and the point, lying exactly opposite in the x direction, on the guide element **29**.
(116) FIG. **22**A and FIG. **22**B show, as examples, two cases in which the guide groove **20** and the guide edge **30** are brought into a parallel course by pivoting about the pivot axis (not shown; the exact position of the pivot axis is unimportant here). In FIG. **21**A, guide groove **20** and guide edge **30** are straight and run parallel to each other at a distance K. The distance J (cf. FIG. **21**A) between

guide elements **29** and tenons **25** should preferably be greater than the distance K.
(117) In FIG. **22**B, the guide groove **20** and the guide edge **30** follow a curved course along the circles K**1** and K**2**, respectively, where the circle K**1** has a smaller radius than the circle K**2**. The circles K**1** and K**2** extend parallel to each other, and the tangents have the distance K from each other at each position. If the distance between the guide elements **29** and tenons **25** were equal to the distance K, the slide element **10** could be moved along the guide groove **20** without there being a pivoting movement of the carrier **28** about the pivot axis D, since the guide elements **29** could also follow the guide edge **30** without resistance.
LIST OF REFERENCE SIGNS
(118) **1** mask **2** mask body **3** face cushion **4** hose attachment **5** forehead support **6** support arm **7** head harness **8** harness clip **9** cushion holder **10** slide element **11** forehead cushion **12** cushion attachment **13** clip holder **14** attachment **15** closure piece **16** connection element **17** rail **18** gas attachment **19** holder frame **20** guide groove **21** closure tab **22** connection **23** connection **24** grip rib **25** tenon **26** slide plate **27** pin **28** carrier **29** guide element **30** guide edge **31** wing element **32** inner wall **33** notch **34** latching lug **35** interspace **36** gap **37** free space **38** cushion receiver **39** peg **40** pocket **41** harness receiver **42** gap **43** interspace **44** subsidiary wing **45** attachment lock **46** wall **47** holder center **48** free space **49** edge **50** guide groove **51** top edge A plane B plane C angle D pivot axis E outer side F inner side G plane of symmetry H distance J distance K distance M distance R**1** radius (guide groove **20**) R**2** radius (guide edge **30**) R**3** radius (guide elements **29**) K**1** circle (guide groove **20**) K**2** circle (guide edge **30**)

Claims

1. A forehead support adjuster for a mask, wherein the forehead support adjuster comprises at least a. a forehead support having a support arm and cushion holder, b. a connection element, c. a rail with at least one guide groove, d. a slide element, the slide element comprising at least one slide plate, at least one tenon and at least one guide element, the at least one tenon being mounted rotatably and slidably in the at least one guide groove of the rail, and the slide element being movably connected to the support arm, and wherein the forehead support adjuster is configured such that a movement of the slide element along the rail results in a pivoting of the forehead support about at least one pivot axis D, and wherein (i) at least two notches are arranged in the rail, and a latching lug is arranged on the at least one tenon, configured so that the latching lug can latch into the at least two notches and thereby fix a pivoting position; and/or (ii) the support arm comprises at least two carriers, which are arranged parallel to each other and are connected to each other via connections which, together with the at least two carriers, form a free space through which the slide element is at least partially plugged; and/or (iii) the support arm transitions at one end into the cushion holder, at an end of the support arm lying opposite the end with the cushion holder a peg being arranged on a connection, via which peg a closure piece is connected captively to the mask, and the cushion holder comprising a holder frame which, together with a holder center, forms a free space into which at least one cushion receiver protrudes from at least one side of the holder frame, in a direction of the holder center, and serves to receive a forehead cushion.
2. The forehead support adjuster of claim 1, wherein at least one inner wall is arranged on the slide plate, the at least one tenon and the at least one guide element being arranged on the inner wall.
3. The forehead support adjuster of claim 1, wherein at least (i) applies.
4. The forehead support adjuster of claim 1, wherein the support arm comprises at least one carrier which is rotatably connected to the connection element via at least one pin.
5. The forehead support adjuster of claim 4, wherein the slide plate has, at two mutually opposite sides, respective wings which are substantially perpendicular to the slide plate and comprise at least in each case two sub-wings, an interspace being present between the respective sub-wings, and wherein discrete markings for marking a pivoting position are arranged on at least one side of the at

least one carrier, wherein the discrete markings are arranged such that at least one marking can be seen in an interspace between the sub-wings when a latching lug fixes a corresponding pivoting position.

6. The forehead support adjuster of claim 1, wherein the rail and the connection element are integrated in a mask body.

7. The forehead support adjuster of claim 1, wherein at least (ii) applies.

8. The forehead support adjuster of claim 7, wherein the carriers of the support arm comprise at least one guide edge, along which the guide elements are guided.

9. The forehead support adjuster of claim 1, wherein at least two inner walls are arranged on the slide plate and are arranged substantially perpendicular to the slide plate, and wherein tenons and guide elements are arranged at edges that lie opposite the slide plate.

10. The forehead support adjuster of claim 1, wherein the guide elements and tenons are arranged at a distance J from each other, and a guide edge and a top edge of the guide groove are at a distance K from each other when the guide edge and the guide groove extend parallel to each other, the distance J being not equal to the distance K and not changing during a movement of the slide element.

11. The forehead support adjuster of claim 10, wherein the guide edge has a curved or straight configuration, a curved guide edge having a radius R**2**, a radius R**1** of the at least guide groove in a curved configuration being not equal to the radius R**2**, and a ratio R**2**:R**1** or R**1**:R**2** ranging from 0.5 to 0.99.

12. The forehead support adjuster of claim 11, wherein the radii R**1** and R**2** range from 7 cm to 15 cm.

13. The forehead support adjuster of claim 1, wherein the at least one guide groove has a curved or straight configuration, a curved guide groove having a radius R**1**.

14. The forehead support adjuster of claim 1, wherein the forehead support can be pivoted by a maximum pivoting angle C of from 10° to 40°.

15. The forehead support adjuster of claim 1, wherein a top edge of the at least one guide groove lies on a circle K**1** with a radius R**1**, and a guide edge lies on a circle K**2** with a radius R**2**.

16. The forehead support adjuster of claim 15, wherein the pivot axis D always has the same relative position, i.e., lying inside or outside, with respect to the circles K**1** and K**2** when a maximum pivoting angle C is not exceeded.

17. The forehead support adjuster of claim 16, wherein the pivot axis D always lies outside the circle K**2** and always inside the circle K**1**.

18. The forehead support adjuster of claim 1, wherein a sliding of the slide element along the rail in a direction of the cushion holder causes the support arm to approach a mask body.

19. The forehead support adjuster of claim 1, wherein at least (iii) applies.

20. A mask, wherein the mask comprises the forehead support adjuster of claim 1 and wherein a gas attachment is arranged between an attachment and the connection element.